Universidad de Oviedo

Escuela Ingenieria Informática

TRABAJO DE FIN DE GRADO
GRADO EN INGENIERÍA INFORMÁTICA DEL SOFTWARE

Evaluación de la influencia de colores en interfaces en modo oscuro mediante eye-tracking

Evaluation of the influence of colors on interfaces in dark mode by eye-tracking

Author: Andrés Cadenas Blanco

Tutor: Dr. Daniel Fernandez Lanvin

Gijón, May 5, 2026

Chapter 1

Acknowledgments

I would like to deeply thank my "unofficial" tutor Sara Vecino García and her incredible patience as well as my tutor Daniel Fernández Lanvin. They have been a guiding light during the whole process and made this thesis possible.

I would like to also thank Lola. Without her support and our random conversations I wouldn't have chosen this topic. Her love along with my family and friends support and care kept me going even in the darkest moments. This work helped me grow as a person and student.

I hope whoever reads this finds it useful and entertaining as I did while writing it.

Kind regards,
Andrés Cadenas Blanco

Chapter 2

Proof of authorship

The Student:	Andrés Cadenas Blanco
With DNI:	53774977B
And UO:	UO282276

DECLARES

That the work presented in this document is the result of his own efforts and that all sources used have been properly cited.

Signed in Gijón, on May 5, 2026

Chapter 3

Abstract

3.1 English

bla bla Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec eleifend mauris erat, sit amet tristique ex lobortis vel. Etiam dapibus tempor nunc, et volutpat libero porta et. Aenean odio dolor, luctus at tincidunt sit amet, egestas sed orci. Praesent nec bibendum arcu, id vulputate velit. Phasellus vel posuere risus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Morbi hendrerit, lectus at lobortis tincidunt, leo nisl convallis lorem, et elementum mi diam rutrum lacus. Phasellus sagittis mauris ac dignissim pretium. Curabitur eu nulla felis. Proin ut iaculis urna. Vivamus sed pellentesque magna. Duis pharetra, lorem eget aliquet ornare, quam dolor lobortis quam, a hendrerit dui eros vitae metus. Interdum et malesuada fames ac ante ipsum primis in faucibus. Pellentesque fermentum leo mi, quis pellentesque magna convallis et.

Vestibulum condimentum tincidunt mauris nec tristique. Donec elementum, turpis id mattis cursus, elit ligula elementum ante, et congue ligula sapien eu neque. Pellentesque feugiat mauris nec consectetur tincidunt. Nunc at quam sit amet elit rhoncus mollis ut euismod velit. Morbi ornare, odio nec faucibus consectetur, quam orci luctus velit, eget pretium ex tellus in quam. Suspendisse eget urna elit. Donec porta sit amet orci posuere varius.

Suspendisse sit amet dui ultricies, tincidunt quam id, faucibus purus. Vestibulum sodales dui in est finibus consequat. Aliquam eget viverra diam. Quisque ac viverra lorem. Donec gravida nulla vel neque aliquam sollicitudin. Ut finibus egestas rhoncus. Aliquam dapibus augue nec risus aliquam, eu euismod nibh pharetra. Mauris ultricies velit diam. Vivamus sed lacus felis.

3.2 Español

Contents

List of Figures

List of Tables

Chapter 4

Introduction

4.1 Context

Computer scientists have always been seen as “hackers” that work with Command Line Interfaces (CLI). This belief comes from a long-gone era where Graphical User Interfaces (GUI) wouldn’t exist. In reality, Command Line Interfaces and Graphical User Interfaces live together in peace in the different sectors of informatics. However, lately the Graphical User Interfaces are experiencing, perhaps what can be called a ‘golden era’. The evolution from the first Graphical User Interface computer, the Xerox Alto in 1973[21],to Web GUIs applications still share the same core. However, they have quickly evolved in the past years thanks to eased access to the Internet and computers. Nowadays, big companies dictate the cutting edge of technology regarding GUIs. A trend that has become obvious is the rise of Dark mode, dividing the whole user community of many applications.

4.2 Motivation

Year after year we have seen how the whole world has changed at a really high pace. Two of the fastest changing areas are Graphical User Interfaces and User Experience (UX). A long time has passed since the times when developers would consider the front end a secondary part of development.

Nowadays a new problem has become viral, not only inside the informatics world but throughout the whole spectrum of internet users: is dark mode better (from now on negative polarity) or bright mode (from now on positive polarity)? This has led many companies to embrace dark mode as its corporative colors, for example the social network X (former Twitter) shifted to negative polarity as its default mode.

There have been many studies regarding usability between both options. However, no one has actually studied the effects of the positive and negative display polarity in longer texts. That has risen the following questions:

- Does the user get more fatigued while reading in negative polarity?
- Will they read faster or slower?

- Will the user struggle to find more information?
- E-commerce websites are mainly programmed in positive polarity. Is the negative polarity really a disadvantage for the user experience while reading and analyzing a product?

All these questions will be analyzed and given the best answer possible in this thesis. A conclusion regarding long texts and display polarities will be studied and explained here.

4.3 Purpose of the thesis

The purpose of the thesis is to investigate the visual fatigue and readability of both display polarities in terms of reading long texts in web-based products. This will be achieved by using a tobii eye-tracker. This tool can give real time data of the eye, giving us objective and solid information.

Visual fatigue can be defined as a moment in which the eyes have a huge workload, and the user can perceive a sensation of discomfort. Visual fatigue can happen after a long period of time in front of a screen, but it also happens every time someone reads a text on a smaller scale. It is usually studied with questionnaires leading to subjectiveness in the analysis. However, in recent times, tools such as eye-trackers have proven a new way to study it. Symptoms of visual fatigue can be calculated as the time the user takes to read a whole paragraph and how many times the person takes to stop at each word.

4.4 Document Structure

The structure of the document has been briefly inspired by Jose Manuel Redondo's spreadsheet of final thesis for development thesis [5]. However, in order to adapt it to an investigation thesis the spreadsheet has been adapted and now doesn't follow non specific standard. This new "version" is mostly inspired by the scientific method by expanding the common topics on a scientific paper, while at the same time being mixed with the most common sections of a usual informatics thesis.

Chapter 5

State of the Art

The state of the art is the information relevant to the study that will be done. In this way, we can close the scope of what this new investigation will prove and find studies that can help us close that new research gap. As a small reminder, display polarity is the mode that will be shown on the screen, positive polarity being dark characters in bright background and negative polarity the other way around.

5.1 Previous Studies

The previous studies section, as explained above, consists of doing research in order to slowly close the research gap. This process was done by following a "baking" process, where an initial research was done on the subject in several Databases such as IEEExplore[2] or Scopus[14], refining the keywords every time to find related articles. The best keywords appeared to be:

- Display polarity
- Visual fatigue
- Reading

After reading and analyzing the first bake, a second bake of refinement and vertical expansion (analyzing cites and references of such scientific papers) was done. After erasing the unrelated articles and the ones out of scope, the following key articles were found:

5.1.1 The impact of web page text-background colour combinations on readability, retention, aesthetics and behavioural intention[4]

Table 5.1: Summary of the article by Hall and Hanna

Authors	Hall, Richard H Hanna, Patrick
Year	2004
Source	Behaviour and Information Technology
DOI	`10.1080/01449290410001669932`

Reason for choice

The motivation to use this paper as a reference is because of the Likert scale questionnaire that is used, giving questions that can perfectly fit with the desired study. In addition, they hand out more questions that could be applied to further explain other non-studied parts of the thesis. They are missing a non preference analysis to further prove their hypothesis, so that is the research gap for this thesis related to this article.

5.1.2 Study on the effects of display color mode and luminance contrast on visual fatigue. [23]

Table 5.2: Summary of the article by Xie *et al.*

Authors	Xie, Xiaojiao Song, Fanghao Liu, Yan Wang, Shurui Yu, Dong
Year	2021
Source	IEE Access
DOI	`10.1109/ACCESS.2021.3061770`

Reason for choice

This study is one of the closest to what the purpose of this thesis is related to. However, there are some differences. For example, the aim is different, and they are not focused on usability but on eye fatigue. Nevertheless, this paper has provided good information to reach the desired experiment for investigating the usability and visual fatigue. It is also important to remark the fact that no questionnaire was used for the evaluation, so there's no preference based evaluation.

In the following illustration we can see the luminance contrast: (a) 0.969, (b) 0.935, (c) 0.855, (d) 0.868, (e) 0.725, (f) 0.469

Figure 5.1: Experimental Illustrations.

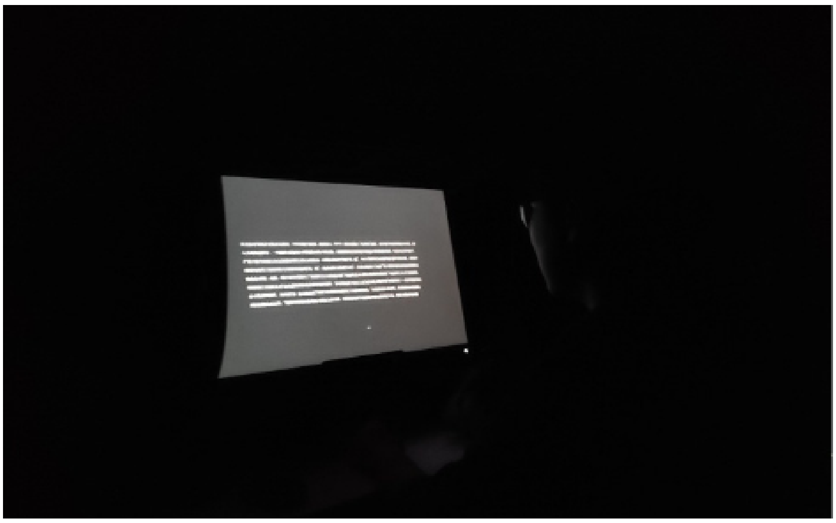

Figure 5.2: Photo of participant during the experiment

5.1.3 The Effect of Display Polarity on Reading Speed and Reading Error Among Young Adults.[9]

Table 5.3: Summary of the article by Muhamad and Mokhtar

Authors	Muhamad, Nurulain Mokhtar, Nurliyana
Year	2024
Source	Journal of Health Science and Medical Research
DOI	`10.31584/jhsmr.20241095`

Reason for choice

The reason for choosing this article is the fact that, even though the study is quite similar, they are not using an eye tracker but instead measuring times by hand, hence giving a general measurement with a lot of imprecisions. However, the methodology (without considering the tools for measuring) is an acceptable one, and some of it is considered while processing the data.

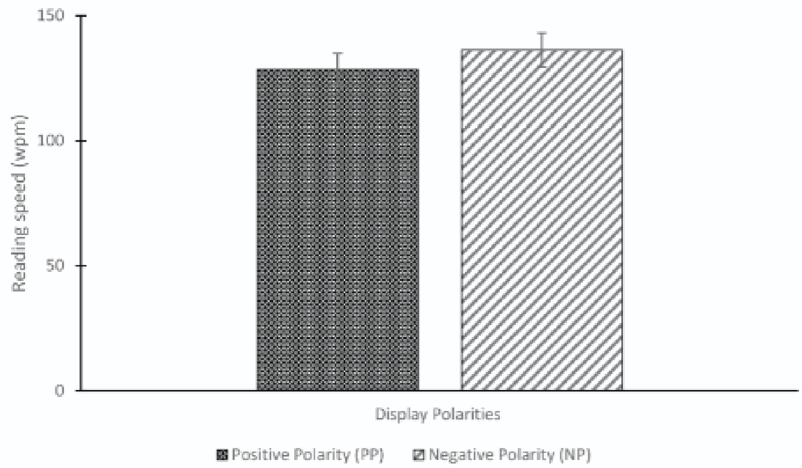

Figure 5.3: The comparison of reading speed between positive and negative display polarities.

In adition to the simmilarities in the methodology, there exists also differences on what they measure. The experimentees are exposed to really long texts and they weren't web based. Therefore only the text was shown. The reading errors were calculating by comapring what the experiment subject read outloud with the original text. This is not at all the objective of this thesis, as it is focused on reading only.

Figure 5.4: The comparison of reading error between positive and negative display polarities.

Chapter 6

Hypothesis

6.1 General Objective

The general objective of this thesis is to evaluate the influence on users reading one of the two main polarities used in modern interfaces. Eventough there are variants of these two main polarities the basic black and white ones were chosen.

Unselected variants of the two main polarities could be:

- Gruvbox: A retro color scheme centered in oranges and greens.
- Evergreen: A greenish colors scheme.
- Nord: A color scheme centered in blues and grays.
- Dracula: A color scheme centered in purples and pinks.

6.2 Specific Objectives

The specific objectives based on the general objective are centered around readability and user preference. Given this experiment is based on eye-tracking long texts in webpages, these two perspectives seems to be the most relevant ones to evaluate. From these two perspectives, the hypothesis are then synthsized as seen in the next section.

6.3 Hypothesis

The hypotheses to be proven in this thesis are:

H1) Users using postive polarity are have more difficulty reading than the negative polarity users

H2) Users using positive polarity read quicker than negative polarity

H3) Users prefer positive polarity over negative polarity

Chapter 7

Methodology

7.1 Experiment Design

The experiment proposal is tightly attached to the idea of using an eye-tracking system, not only because it is an innovation that gives us more precision within the measurements but it also uniqueness surrounding the state of the art as really few experiments have used this device. To be more exact we will be using the eye tracking model Tobii Pro Nano[19]. The experiment is briefly described in this section, but it will be further explained in the chapter 8.

The user will be prompted with a text message showing the instructions. This text comes along with a previous tobii calibration that will be further explained in the section 8.2.1 and 8.2.2 respectively.

After the instructions and calibration are completed, the second phase, the actual experiment can start. There are two possible paths, negative display polarity path or positive display polarity path. Both affect the instruction texts as long as the screenshots shown to the user. The wiki text was shown always the first being followed by the product description. The user has as much time as they want. However, it wasn't possible to return to previous pages.

Once the user finished reading the third phase would start, a questionnaire regarding preference will be shown, it also contains a small set of questions about the texts to ensure the user has been reading carefully.

In addition, the webpages chosen for the experiment are `wikipedia.org` and `amazon.es`. These two webpages will be further explained in the 8.2.3 section.

7.2 Population of Study

In order to have a good experiment, not only is it important to have a consistent and well-prepared process but also have a focus population of study. Therefore, the objective age was young persons between 18 to 30 years old. Before making the experiment the experiment subjects had to sign a consent form of being subject of the experiment and to treat their data.

It is also worth mentioning that the objective was to find the usual user on the internet. For that an average user isn't a computer scientist as they can be specialized on web design or other similar fields. This point made us go to the faculty of Economics as it was a larger faculty and the students can be considered an "average" user.

The population of the study finally consisted of 34 persons, divided evenly between the negative and the positive polarity. Half of the population were women and the other half were men. In adition, the corrected vision of the users was also evenly distributed as seen in chapter 10. This population is enough to ensure data consistency, however as explained in the chapter 12 it is interesting to increase the population in the future to gain even more consistency for the scientific paper.

To summarize, it can be concluded that the conditions and population of the study were:

Table 7.1: Distribution of participants by gender, polarity, and age.

Gender	**Negative Polarity**	**Positive Polarity**	**Total**	**Age**
Men	7	7	14	[18,30]
Women	10	10	20	[18,30]

7.3 Tools

In order to accomplish the success of the experiment there is a need of some tools. They will be further explained why they are important ant what purpose do they serve. There is one hardware tool requirement and 3 software tools that are used to collect data and process it.

7.3.1 Tobii Pro Nano

The tobii pro nano it's an eye-tracker currently discontinued by the owners (Tobii)[19] that is still to its day a great tool to collect user data. It's worth mentioning that this tool is quite expensive and to use it the University of Oviedo has lent me one unit. Always with the surveillance of my tutors. The tobii pro nano is the lightest and most portable unit offered, however it has some limitations that won't be a problem in our case as the limitations are for precise operations.

Figure 7.1: Tobii Pro Nano Device[19]

Eye Tracker Technology Breakdown

The eye tracker works in an amazing way, the process is a combination of high-quality cameras. Using what's commonly called Pupil Center Corneal Reflection (PCCR). In the case of tobii pro nano it uses a patented variant of PCCR (US Patent US7,572,008).[17]

The PCCR is a mix between "illuminators" that is components that are almost infrared sensors that create such light on the eyes. Then different sets of cameras take pictures of the eye patterns and reflections. That is sent to algorithms that detect those patterns and reflections creating a detailed 3D model of the eye position and its gazes [17].

7.3.2 Tobii Pro Lab

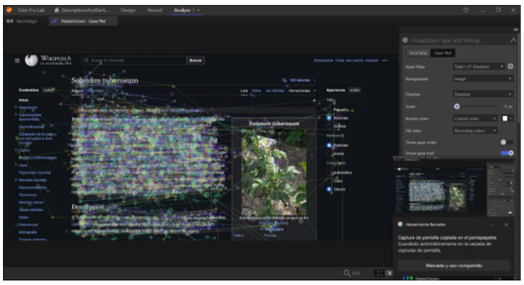

Figure 7.2: Tobii Pro Lab Software[18]

The eye-tracking device connects to a software application called "Tobii Pro Lab". This application aims to ease the use, collection and analysis of the gaze data. Curiously this is a negative display polarity program. It is needed to have a license in order to use it. However, a 30 day trial is provided which has been enough toextract the data, change or create the desired data, download it and process it. The data was gathered in the laptop of my proffesor as they have the paid license.Some of the most interesting tools are the visualization of heat maps, recordings and the Areas of Interest, that will be further explained.

7.3.3 Google Forms

Google Forms is a free tool developed by Google that allows google users to create forms. This is a powerful tool as the answers can be stored in a Google Spreadsheet that can be exported as a .csv archive to be later processed. This tool is also pretty compatible with the tobii pro lab when creating the timelines that the user is going to be exposed due to the fact of the possibility of running.

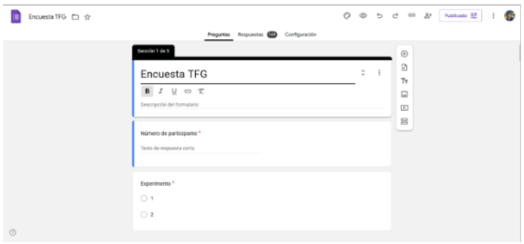

Figure 7.3: Google Forms Example

7.3.4 Python 3

There're few programming languages that almost everyone on earth has heard of, this is the case. Python is a powerful programming language not because of its speed or performance but because of the libraries that are available. This python environment makes room to one of the biggest spaces for data processing. In my case, I'm not a specialized student on data processing, that's why a well-known and big space like this can be the perfect tool to approach a thesis.

Python external libraries

As mentioned beforehand, the prowess of python resides in its libraries. The libraries used for this thesis are:

- **Pandas**[11]: Pandas is a library centered around the idea of easing data manipulation and analysis by providing a generic data structure called DataFrame. It is almost an standard in the data ecosystem of python.

- **Matplotlib**[8]: Matplotlib strength resides in its flexibility to create any kind of graph. It is one of the most used libraries in Python.
- **NumPy**[10]: NumPy name comes from "Numerical Python" and it is a library that provides aid for the manipulation of arrays of data. It combines pretty well with Pandas as every column or row of a DataFrame can be transformed into a NumPy array easily.
- **SciPy**[13]: SciPy is a library created for scientific and technical computing. Providing a large amount of functions for many different fields. In the case of this study, the biggest use resided in the statistics algorithms like Mann-Whitney U test and the t-test.

7.3.5 Dark Reader

Given that the objective is to analyze the effects of display polarity on texts the decision was to choose a wiki page as it is a tool that until this day many users do. Moreover, an e-commerce is another safe bet. In order to make it the most average the decision was to use "wikiedia.org" and "amazon.es". However, the e-commerce website didn't allow negative polarity, so we had to use a firefox extension called Dark Reader which you can see in figure 7.4

Figure 7.4: Dark Reader Extension[1]

Chapter 8

Experiment Execution

8.1 Environmental Conditions

The experiment was developed in the Faculty of Economics of the University of Oviedo, located in the city of Oviedo during the 24th and 25th of April. The weather was cloudy with few intermittent rains, this led to a general dim light. In order to have proper lighting and stable conditions the room was illuminated with fluorescent lights giving a total oscillation of 7,2 to 7,3 Exposure Value (EV).

Regarding the placement of the experiment, we were in a Main Hall in the calmest possible place. Experiment subjects would sit in a chair with a small table holding the laptop and the eye-tracker. The average distance between the experiment subjects and the screen was 60 cm as suggested by the tobii pro lab.

8.2 Experiment Phases

The first and most 3 phase of the experiment consisted of a tobii calibration. As this is a key factor and it was desired to have a lot of precision during the experiment, that is why the 9-point calibration was chosen. Experiment subjects were asked to watch and follow a white dot on the screen after this was finished the calibration results were shown. In the case the results were favorable the experiment would continue or else the calibration would be repeated entirely or partially.

8.2.1 Eye-tracker Calibration

To begin the experiment, the subject was asked to sit in front of the screen and told how to navigate through the experiment. After that, the calibration of the eye-tracker starts, this calibration is done following a dot through the screen. There are different versions of such calibration and the 9-point calibration was chosen as it is the most precise one. The following figure 8.1 is from an unrelated scientific paper called "Understanding how low vision people read using eye tracking"[20] but it is relevant for the calibration as it shows the differenc calibration options.

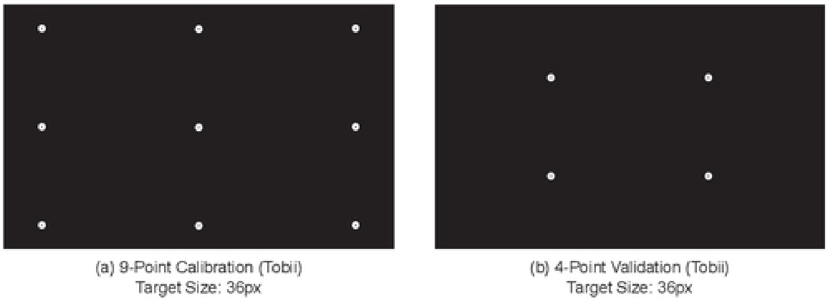

Figure 8.1: Different point calibrations for tobii pro nano, from

8.2.2 User Instructions

The second phase of the experiment already had two different versions. One consisted of the instructions displayed in negative polarity and the other one prompted the instructions in positive polarity. Results of this have been recorded but not taken into account inside the experiment as they are outside of the aim of web based texts.

Figure 8.2: Instructions in positive and negative display polarity

8.2.3 Reading Task

The reading task consisted of 2 texts. The first texts is based on Wikipedia newest negative and positive display polarity while the first text is based on Amazon web-page along with the plugin "Dark Reader" explained above in the document. The pages displayed were centered around daily objects with complex enough information so It wouldn't be a technical hurdle while not being too simple to read.

Wikipedia article

The chosen Wikipedia page regards potatoes[22] as this is a common object with complex enough information so the subjects wouldn't have any problem reading it but at the same time it doesn't contain technical knowledge. The screenshots of the page in both polarities can be seen in the figures 8.3 and 8.4. The native negative polarity mode of Wikipedia was used in the test.

Figure 8.3: Wikipedia page about potatoes in positive display polarity[22]

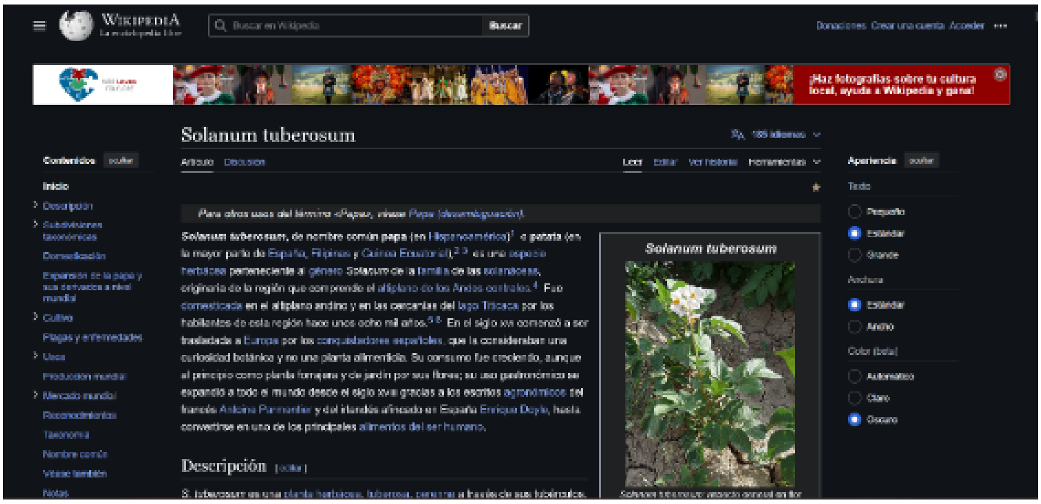

Figure 8.4: Wikipedia page about potatoes in negative display polarity[22]

Amazon product

The second webpage chosen was the e-commerce webpage of Amazon. This provides a broad variety of possibilities. The product was chosen arbitrarily ending up with a backpack[6] as it is a common object and the information provided was complex and wit long descriptions. The screenshots of the page in both polarities can be seen in the figures 8.5 and 8.6. Being 8.5 the positive display polarity and 8.6 the negative display using the Dark Reader extension[1] that was explained in 7 chapter.

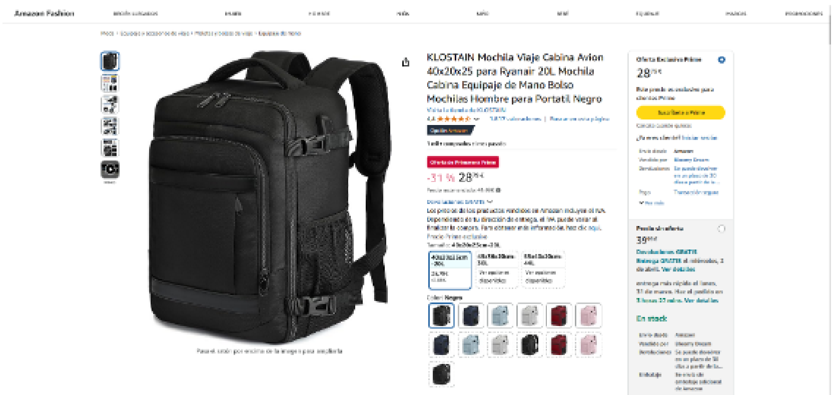

Figure 8.5: Amazon product page in positive display polarity[6]

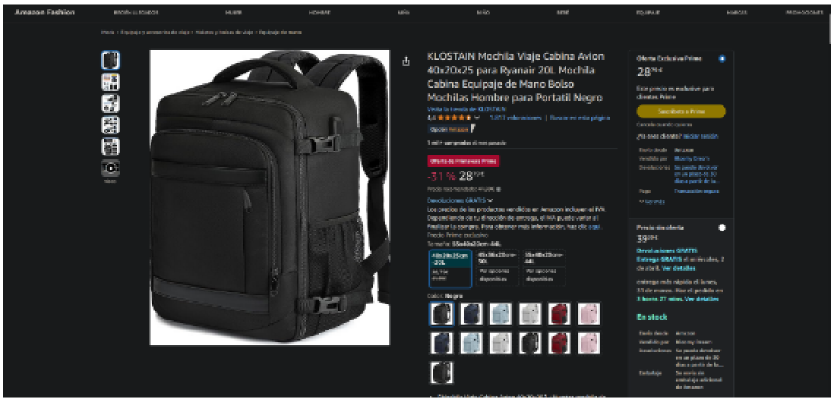

Figure 8.6: Amazon product page in negative display polarity[6]

8.2.4 Final Questionnaire

The final phase of the experiment consisted of a questionnaire which had 5 different sections:

1. Identification information.
2. Personal data.
3. Wikibased questions.
4. E-commerce based questions.
5. Preference and subjective questions.

Identification information

This section was a key part as it allowed for the linking of the results with the eye-tracking data. These two first questions were:

- **Subject number:** This number was provided to the subjects.
- **Experiment:** Experiment 1 or Experiment 2 depending on the display polarity. This was also provided to the subjects.

Personal Data

These questions were the only two personal data questions that were gathered during the experiment. The objective of them was to have control over the experiment population and to be able to check for differences between different groups. The questions were:

- **Gender:** Male, Female or Other.
- **Age:** In format dd-mm-yyyy.

Wiki-based questions

The objective of this section was to check the reading comprehension of the subjects regarding the article. The questions were:

1. **What is the gender of the potato?**

   Solanum (Correct)

   Tuberosa

   Herbacea

2. **Which French person expanded the gastronomic use of potatoes?**

   Paul Leduc

   Antoine Parmentier (Correct)

   Alan Pinaud

3. **What was the original purpose of the potato?**

   Healing plant

   Gastronomic use

   Garden decoration (Correct)

4. **In which lake was the potato first domesticated?**

   Titicaca (Correct)

   Maracaibo

   Cocibolca

E-commerce based questions

The e-commerce reading questions aimed to be simmilar to the wiki-based ones. Whoever, there was a question that had significant impact depending on display polarity.

The question that had this conflict asked the color of the backpack, and subjects in negative display polarity had a tendency to answer that the backpack was dark grey rather than black. This will be further explained in the 12 chapter.

The questions asked in this section were:

1. **Does this product comply with Ryanair restrictions?**

   Yes (Correct)

   No

2. **What is the capacity of the backpack?**

   20L (Correct)

   30L

   40L

3. **What is the color of the backpack?**

   White

   Black (Correct)

   Dark grey

4. **Which is the purpose of the isolated compartment in the main part?**

   To carry a laptop

   To carry wet objects (Correct)

Preference and subjective questions

The final section of the questionnaire collected personal preferences that are not going to be directly analyzed in the experiment. However, future work can be directed towards the analysis of these preferences and their relation with the results of this thesis. The first three questions were a multiple choice and then the other ones were likert scale of 5 points.

Most of the questions here were adapted from the ones in the papers "The impact of web page text-background colour combinations on readability, retention, aesthetics and behavioural intention"[4] and "Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture" [3] both articles center around a likert scale questionnaire based on preference. The first article is centered around display polarity and the second one about webpage preference in general. Both were adapted to fit the needs of this thesis.

These questions were:

1. **What is your type of vision?**

   Normal vision (No need for correction)

Corrected vision (Glasses or contact lenses)

Other (Please specify)

2. **Which mode do you prefer for wiki-based websites?**

   Bright mode

   Dark mode

3. **Which mode do you prefer for e-commerce websites?**

   Bright mode

   Dark mode

4. **In general, how frequently do you use bright mode?**
5. **In general, how frequently do you use dark mode?**
6. **It is easy to read the text with the font and colors used**
7. **The colors used in the webpage are pleasant**
8. **In general, I'm happy with the webpage graphical interface**
9. **The color combinations made the reading experience better**
10. **The color combinations made the memorization of the information better**
11. **The color combinations were pleasent at sight**
12. **The color combinations looked professional**

Chapter 9

Data Analysis

9.1 Data Preprocessing

Data preprocessing is important for every investigation as it prevents the poisoning of the results with undesired outliers. In this case, there are two main sourrces of data, the tobii data and the google forms questionnaire. The data preprocessing included heavy manage of .tsv files and .csv files.

9.1.1 Tobii Data

The tobii data was preprocessed using the tool Tobii Pro Lab seen in chapter 7. The process consisted of importing the .tsv files and creating a pandas dataframe that would later be divided into two different dataframes. The first dataframe would be the positive display polarity and the second dataframe would be the negative display polarity. This is possible thanks to the common column to all exported data called "Timeline". This specific column showed wich experiment was being performed at the moment of the recording.

This separation was later divided for the different scenarios: Wikipedia webpage and Amazon webpage. This is achieved thanks to the TOI (Time of Interest) column, which showed the scenes of the experiment.

```
def filter_by_polarity(df):
    '''
    Function to filter the DataFrame by polarity (claro
       and oscuro)

    There must be a column named Timeline with the
       values 'Timeline1-Mode1' for claro and 'Timeline2
       -Mode2' for oscuro
    '''
    claro = df[df['Timeline'] == 'Timeline1-Mode1']
    oscuro = df[df['Timeline'] == 'Timeline2-Mode2']
    return claro, oscuro
def separate_by_toi(df, toi):
    '''
    Function to obtain a given TOI name
```

```
    There must be a column named 'TOI'
    '''
    df_toi = df[df['TOI'] == toi]
    return df_toi
tf = pd.read_csv('raw/TotalFixations.tsv', sep='\t')

tf_claro, tf_oscuro = filter_by_polarity(tf)


tf_claro_patata = separate_by_toi(tf_claro, 'PatataClaro
   ')
tf_oscuro_patata = separate_by_toi(tf_oscuro, '
   PatataOscuro')

tf_claro_mochila = separate_by_toi(tf_claro, '
   MochilaClaro')
tf_oscuro_mochila = separate_by_toi(tf_oscuro, '
   MochilaOscuro')
```

The data outliers were not taken into account for the analysis, as they were not considered to be a problem for the results and there were really few of them. Therefore no more preprocessing was needed for the tobii data.

9.1.2 Google Forms Data

The google forms data needed much preprocessing due to the fact of human intervention and a bug related to Tobii Pro Lab. Firstly the data was imported as a pandas dataframe from the CSV file downloaded from the google spreadsheet generated for the form.

The second step was to declutter the dataframe by removing all the duplicated rows. These duplicated rows are generated by a bug in tobii pro lab that makes the explorer to send many requests rather than one, which ended up in generating an undefined number of duplicates per person. This was solved with the following code. Note that due to Latex nature the original column contains an accent in the "u" of número but the code blocks do not allow it so it was changed to avoid the compiling error.

```
    df = pd.read_csv('raw/Encuesta TFG.csv')
    df = df.drop_duplicates(subset=['Numero de
       participante'], keep='first').copy()
```

Next, the age column needed detailed work as some people set their age and some others their birth date. This was solved with a function that checks if the value is a number, in such case it returns the number converted to an integer. In the other case, it tries to parse the date with the pandas function ‘to_datetime()‘ this can result in a warning but it is not an issue. The warning explains that when using the days first format, a date is introduced starting with something else than the day.

```
# Convertimos el string a fecha
df['Marca temporal'] = pd.to_datetime(df['Marca temporal
    '], dayfirst=True, errors='coerce')

def calculate_age(age_value, ref_ts):
    # Hay gente que puso la edad numerica solo
    age_num = pd.to_numeric(str(age_value).strip(),
        errors='coerce')
    if pd.notna(age_num):
        return int(age_num)

   # si puso fecha, la calculamos y comparamos con la
      fecha de referencia del csv (ref_ts)
   # Esto genera un warning pero da igual porq no es
      nada que afecte el codigo
    birth_ts = pd.to_datetime(str(age_value).strip(),
        dayfirst=True, errors='coerce')
    if pd.notna(birth_ts) and pd.notna(ref_ts):
        years = ref_ts.year - birth_ts.year - ((ref_ts.
            month, ref_ts.day) < (birth_ts.month,
            birth_ts.day))
        return int(years)

# Llamamos a la funcion
df['Edad'] = [calculate_age(v, t) for v, t in zip(df['
    Edad'], df['Marca temporal'])]
```

In the case of the date being parsed, then the formula to calculate the age is the following:

$$Age = \text{Current Year} - \text{Birth Year} - (1 \text{ or } 0) \tag{9.1}$$

The final step of 1 or 0 checks whether the person has already had their birthday this year or not, in the case of having had the birthday, then the age is not reduced, otherwise it is reduced one year. After computing all the ages one subject did not provide the correct age information but rather gave the actual year.

Finally the last step was to preprocess the question regarding the type of vision of the subjects. This was done by hand as there were some unique cases that wouldn't make sense to be solved programming. This question will be further examined in the chapter 10.

9.2 Chosen Metrics

The chosen metrics to analyze are based on the article "Eye tracking in human-computer interaction and usability research: Current status and future prospects"[12]. These metrics are based on the eye-tracking data and were analyzed for the different escenarios as long as with the whole recording.

9.2.1 Fixations

Fixations are the moments when the eye stays still focused on a point of the screen. This moments can be as small as a few milliseconds. There are a few metrics which are related to fixations as shown below with their respective definition by Poole, Alex and Ball, Linden J.[12]

Table 9.1: Fixation Metrics and their corresponding variables in the .tsv file

Fixation Metric	**Variable in .tsv file**
Gaze	Total_duration_of_whole_fixations
Total Number of Fixations	Number_of_whole_fixations
Duration of First Fixation	Duration_of_first_whole_fixation
Average Duration of Fixations	Average_duration_of_whole_fixations

- **Gaze (sum of the duration of all fixations):** The Gaze is the result of all the fixations duration in a prescribed area. It is best used to compare attention distributed between targets. It can also be used as a measure of anticipation in situation awareness if longer gazes fall on an area of interest before a possible event occurring. [12]
  - Although this metric is centered in Areas of Interest, in this thesis the 2 areas compared are positive display polarity and negative display polarity.
- **Total Number of Fixations:** More fixations overall results in less efficient reading.
- **Duration of Fixation:** When a fixation takes more time it can be an indicator of difficulty to process the data or the user being interested in its content.

- This will be analyzed both with the first fixation and with the average duration of the fixations.

9.2.2 Saccades

Saccades are the time the eye is moving from one fixation to another, during this period the brain is not processing any information. They can provide information that supports the fixations or that can be useful in this thesis. These metrics will be explained briefly with the definitions given by Poole, Alex and Ball, Linden J.[12]

- **Saccade Amplitue:** Saccades usually are small so bigger saccades mean that the user attention is being drawn
  - The Saccade Amplitude is analyze through the average amplitude, as well as the average peak velocity of the saccades.
- **Peak Velocity:** It corresponds to the maximum velocity of the saccade and it can be an indicator of fatigue.
- **Number of Saccades:** More saccades means more searching and more regressions, that will be explained later.

Table 9.2: Saccade Metrics and their corresponding variables in the .tsv file

Saccade Metric	Variable in .tsv file
Saccade Velocity	Average_peak_velocity_of_saccades
Average Saccade Amplitude	Average_amplitude_of_saccades
Amplitude of First Saccade	Amplitude_of_first_saccade
Number of Saccades	Number_of_saccades

Regressions

Regressions are the saccades that go back to a previous part of the text. Usually the regressions are small and only go back a few letters [12] but they can also be big and go back many words or even paragraphs. In order to analyze this metric. The chosen strategy resides in counting the number of fixations inside the text areas of interest, theoretically, a perfect reading would have one fixation per word, therefore, if there are more fixations it means that there are more regressions. It is not the most precise way but it should give a good approximation for this thesis.

9.3 Statistical Analysis

In this section the process of analyzing the data with statistical methods will be further explained. This process consists of 3 main steps that are shared along all the studied data. Please, keep in mind that all three statistical methods have the α value of 0,05 but relevant results under 0,1 will also be recorded. In adition, no extense explanation of the method will be given as this is out of the scope of the thesis.

1. **Extract Basic Data:** Firstly the standard deviation, mean, maximum and minimum of each group is extracted to obtain the general shape. It will be later used in the representation of the results of the hypothesis.

2. **Shapiro-Wilk Test[15]:** Secondly the data is analyzed with the Shapiro-Wilk test to check the normality of it. The null hypothesis of this test proves that the data follows a normal distribution, from now on, this statistic will be called W. In the following chapters W_{PP} and W_{NP} will represent the Shapiro-Wilk test for Positive Polarity and Negative Polarity respectively.

$$W = \frac{(\sum_{i=1}^{n} a_i x_{(i)})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

3. **T-Student test[16] or Mann-Whitney U[7] test** Finally, a comparison of both groups W is compared. In the case of both datasets following a normal distribution, then the T-Student test is used, otherwise the Mann-Whitney U test is used. Note that in both cases the α value is 0,05 but relevant results under 0,1 will also be recorded.

   T-Student test: The T-Student test compares the means of two normall distributed groups. Being the null hypothesis that both groups have the same mean. Therefore it helps to check whether the difference between the groups is significant or not. In this study, the chosen variation is the independent samples test.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

   Mann-Whitney U test: The Mann-Whitney U test is a non-parametric test which compares the medians of two samples being the null hypothesis that both groups have the same median. Altough it is different from the T-Student results should be simmilar and it also provides a way to test data that is not normally distributed.

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

Chapter 10

Results

10.1 Eye-traker Data

The study developed in python has proven interesting results regarding the metrics mentioned in chapter 9. The results will show the results of the statistics along with the basic statistical values such as mean and standard deviation.

10.1.1 Total Duration of Whole Fixations

Table 10.1 shows that the mean was higher in length for positive polarity in all TOIs, having the biggest difference in the Wiki-Based Scene.

Table 10.1: Results of Total Duration of Whole Fixations

Display Polarity	**TOI**	**Min. (ms)**	**Mean (ms)**	**Std. Deviation (ms)**	**Max. (ms)**
Positive Polarity	**Wiki-based**	18570	52902.88	24410.26	125026
	E-Commerce	9443	64217.12	34024.20	125642
	Entire Exp.	215011	347509	81449.08	504168
Negative Polarity	**Wiki-based**	2631	41879.59	17204.72	72631
	E-Commerce	8977	53989.41	22480.35	94915
	Entire Exp.	154088	282626.65	63197.95	391933

Figure 10.1 shows how only the entire experiment has significant results within this variable. Further statistics can be seen in table 10.2 where p-values under $\alpha = 0.05$ are highlighted in red and p-values under $\alpha = 0.1$ are highlighted in orange.

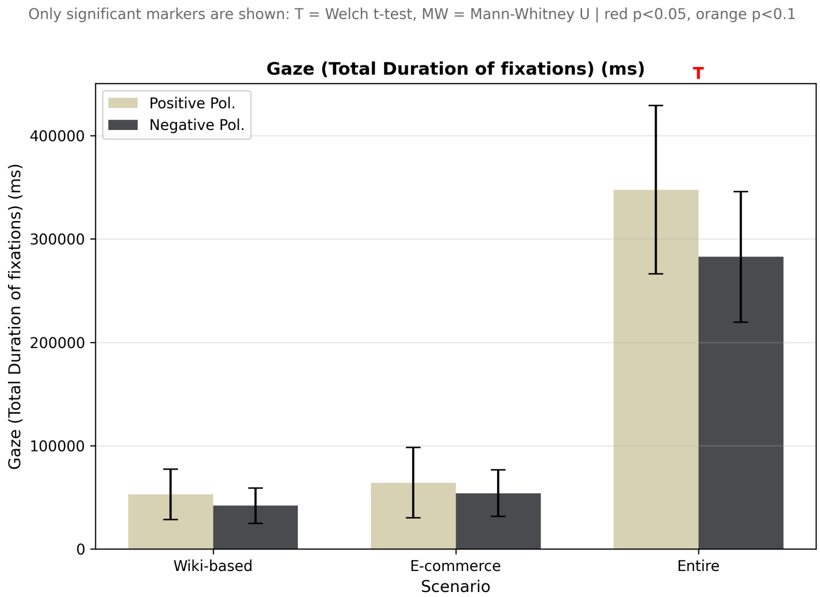

Figure 10.1: Total Duration of Fixations for each Display Polarity and TOI.

Table 10.2: Statistics of Total Duration of Whole Fixations

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	**0.0091**	0.7774	-	0.2557
E-Commerce	0.7094	0.9138	0.3244	-
Entire Exp.	0.8348	0.9790	**0.0174**	-

10.1.2 Total Number of Fixations

Regarding the total number of fixations, table 10.3 proves that the mean number of fixations is higher in positive polarity in every single TOI.

Table 10.3: Results of Total Number of Fixations

Display Polarity	TOI	Min.	Mean	Std. Deviation	Max.
Positive Polarity	**Wiki-based**	62	158.94	52.41	301
	E-Commerce	35	199.06	92.76	320
	Entire Exp.	545	979.59	240.28	1510
Negative Polarity	**Wiki-based**	13	142.65	50.59	208
	E-Commerce	45	194.06	64.46	280
	Entire Exp.	405	844.82	162.51	1159

The statistical analysis, as seen in figure 10.2 and table 10.4. There is no significant difference between both groups but the entire experiment is close to be significant with a p-value of 0.0736 in the T-Student test.

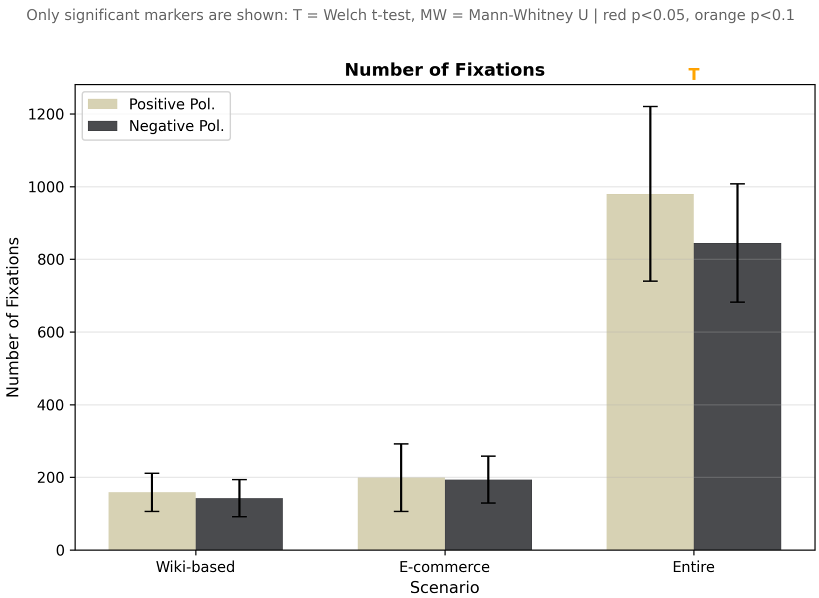

Figure 10.2: Total Number of Fixations for each Display Polarity and TOI.

Table 10.4: Statistics of Total Number of Fixations

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	0.5150	0.2003	0.3776	-
E-Commerce	0.1019	0.1089	0.8591	-
Entire Exp.	0.5321	0.3570	**0.0736**	-

10.1.3 Duration of First Fixation

The first duration of fixation is lower in positive polarity in both TOIS but differs in the whole recording, as shown in table 10.5.

Table 10.5: Results of Duration of First Fixation

Display Polarity	TOI	Min. (ms)	Mean (ms)	Std. Deviation (ms)	Max. (ms)
Positive Polarity	**Wiki-based**	83	181.21	111.99	583
	E-Commerce	133	193	54.03	316
	Entire Exp.	67	315.47	268.45	1016
Negative Polarity	**Wiki-based**	100	205.82	100.07	533
	E-Commerce	83	200	123.55	650
	Entire Exp.	67	203.88	163.85	766

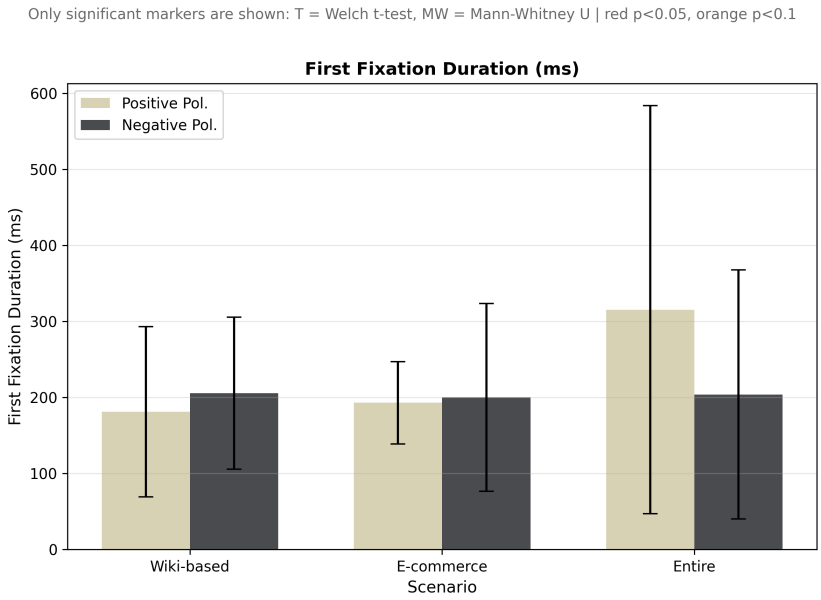

Figure 10.3: Duration of First Fixation for each Display Polarity and TOI.

The statistical analysis in this case shows how none of the TOIS follows a normal distribution, as shown in table 10.6 resulting in the Mann-Whitney U test where none of the TOIS is significant, see figure 10.3.

Table 10.6: Statistics of Duration of First Fixation

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	**0.0001**	**0.0005**	-	0.3224
E-Commerce	**0.0048**	**0.0000**	-	0.5434
Entire Exp.	**0.0046**	**0.0002**	-	0.3005

10.1.4 Average Duration of Fixations

The average duration of fixations seems to be higher in positive polarity in every single TOI, as shown in table 10.7. This difference is more steep in the Wiki-Based scene.

Table 10.7: Results of Average Duration of Fixations

Display Polarity	TOI	Min. (ms)	Mean (ms)	Std. Deviation (ms)	Max. (ms)
Positive Polarity	**Wiki-based**	234	329.29	79.39	537
	E-Commerce	236	318.41	63.58	457
	Entire Exp.	288	358.88	43.08	451
Negative Polarity	**Wiki-based**	202	287.53	57.97	441
	E-Commerce	193	272.19	58.81	407
	Entire Exp.	262	336.18	46.65	436

Regarding the statistical analysis, only a relevant result exists for the E-Commerce scene with the p-value of 0.0514 being pretty close to the significance level.

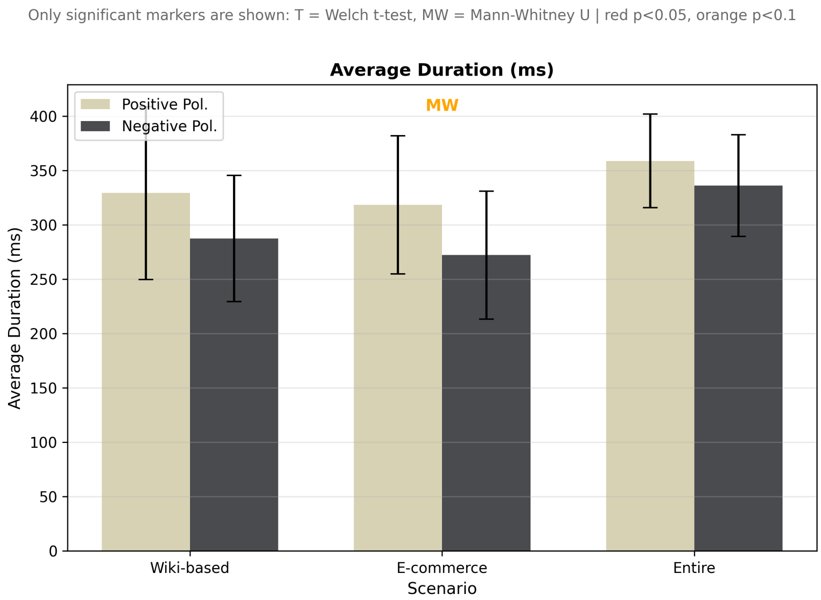

Figure 10.4: Average Duration of Fixations for each Display Polarity and TOI.

Table 10.8: Statistics of Average Duration of Fixations

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	**0.0365**	0.2901	-	0.1528
E-Commerce	**0.0200**	0.4693	-	**0.0514**
Entire Exp.	0.6286	0.8766	0.1624	-

10.1.5 Average Peak Velocity

The average peak velocity is higher in negative polarity for every TOI as seen in table 10.9. However, It is only relevant in the entire recording with a p-value of 0.0871 proven in table 10.10 and figure 10.5

Table 10.9: Results of Average Peak Velocity

Display Polarity	TOI	Min. (º/s)	Mean (º/s)	Std. Deviation (º/s)	Max. (º/s)
Positive Polarity	**Wiki-based**	89.13	107.76	15.84	148.82
	E-Commerce	77.19	102.25	24.92	189.92
	Entire Exp.	91.69	106.49	7.09	119.99
Negative Polarity	**Wiki-based**	79.57	118.72	29.61	216.31
	E-Commerce	76.66	104.02	16.63	133.35
	Entire Exp.	88.48	112.61	11.79	135.44

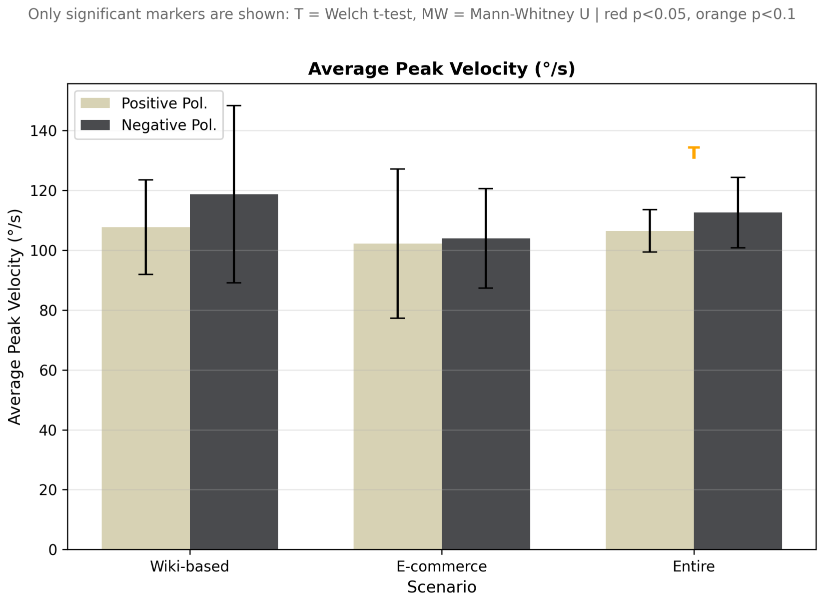

Figure 10.5: Average Saccade Peak Velocity for each Display Polarity and TOI.

Table 10.10: Statistics of Average Peak Velocity

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	**0.0371**	**0.0030**	-	0.2557
E-Commerce	**0.0001**	0.5860	-	0.3892
Entire Exp.	0.3574	0.9985	0.0871	-

10.1.6 Average Saccade Amplitude

The results of table 10.11 show a pretty even mean distribution along polarities making no evident differences. This can be further seen in figure 10.6 and table 10.12 where none of the statistics is close to be significant.

Table 10.11: Results of Average Saccade Amplitude

Display Polarity	TOI	Min. (º)	Mean (º)	Std. Deviation (º)	Max. (º)
Positive Polarity	**Wiki-based**	2.22	2.97	0.48	3.87
	E-Commerce	1.92	2.95	0.70	5.15
	Entire Exp.	2.39	3.12	0.36	3.70
Negative Polarity	**Wiki-based**	2.26	3.26	0.69	5.30
	E-Commerce	2.14	2.98	0.50	4.06
	Entire Exp.	2.63	3.31	0.39	4.10

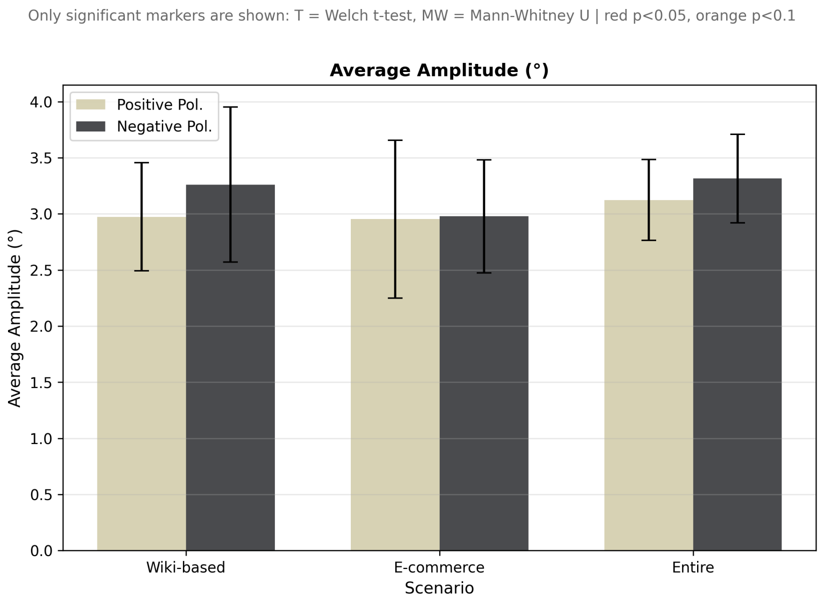

Figure 10.6: Average Saccade Amplitude for each Display Polarity and TOI.

Table 10.12: Statistics of Average Saccade Amplitude

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	0.6837	**0.0758**	0.1833	-
E-Commerce	**0.0153**	0.6737	-	0.6297
Entire Exp.	**0.0984**	0.9634	0.1620	-

10.1.7 Amplitude of First Saccade

The amplitude of the first saccade behaved simmilarly to the average saccade amplitude with a pretty even mean distribution. Followd with a big standard deviation shown in table 10.13. This can be further seen in figure 10.7 and table 10.14 this also leads to no significant nor meaningful results in the statistics table 10.14

Table 10.13: Results of Amplitude of First Saccade

Display Polarity	**TOI**	**Min. (º)**	**Mean (º)**	**Std. Deviation (º)**	**Max. (º)**
Positive Polarity	**Wiki-based**	1.59	9.06	3.93	17.71
	E-Commerce	0.56	5.35	3.29	13.00
	Entire Exp.	1.07	5.18	3.97	17.74
Negative Polarity	**Wiki-based**	2.03	8.74	3.79	14.09
	E-Commerce	0.56	4.70	3.39	14.86
	Entire Exp.	0.59	6.46	4.57	17.47

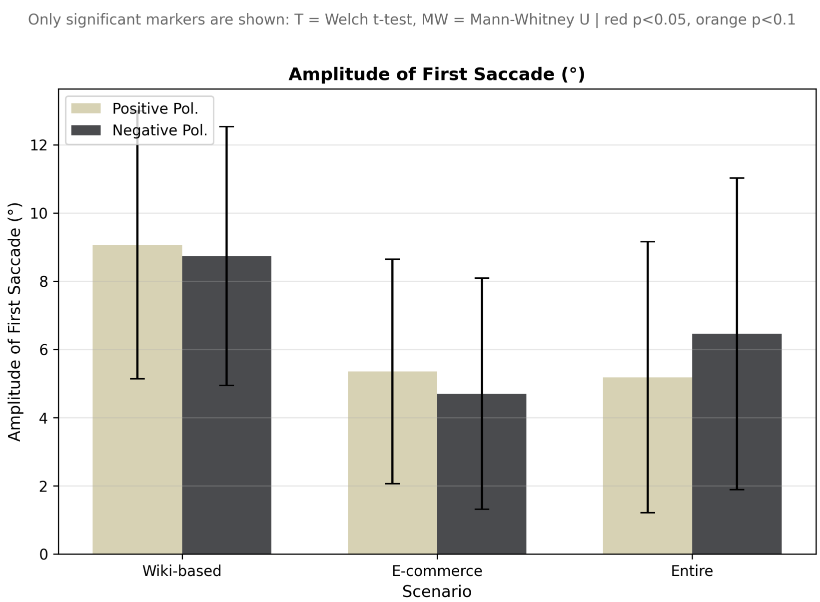

Figure 10.7: Amplitude of First Saccade for each Display Polarity and TOI.

Table 10.14: Statistics of Amplitude of First Saccade

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	0.5576	0.2155	0.8151	-
E-Commerce	0.3231	**0.0159**	-	0.5018
Entire Exp.	**0.0030**	0.1831	-	0.5353

10.1.8 Number of Saccades

Lastly, all 3 scenes had a higher mean number of saccades in positive polarity than in negative polarity clearly shown in table 10.15.

Table 10.15: Results of Number of Saccades

Display Polarity	TOI	Min.	Mean	Std. Deviation	Max.
Positive Polarity	**Wiki-based**	59	151.29	47.86	270
	E-Commerce	34	184.18	83.67	305
	Entire Exp.	509	870.06	188.81	1122
Negative Polarity	**Wiki-based**	13	134.41	50.24	199
	E-Commerce	29	171.47	59.10	241
	Entire Exp.	344	746.41	166.22	1004

There's only a relevant results regardin the whole recording with a p-value of 0.0582 in the t-student test. It is not enough to be considered relevant but gets close enough to be considered relevant. This can be further seen in figure 10.8 and table 10.16.

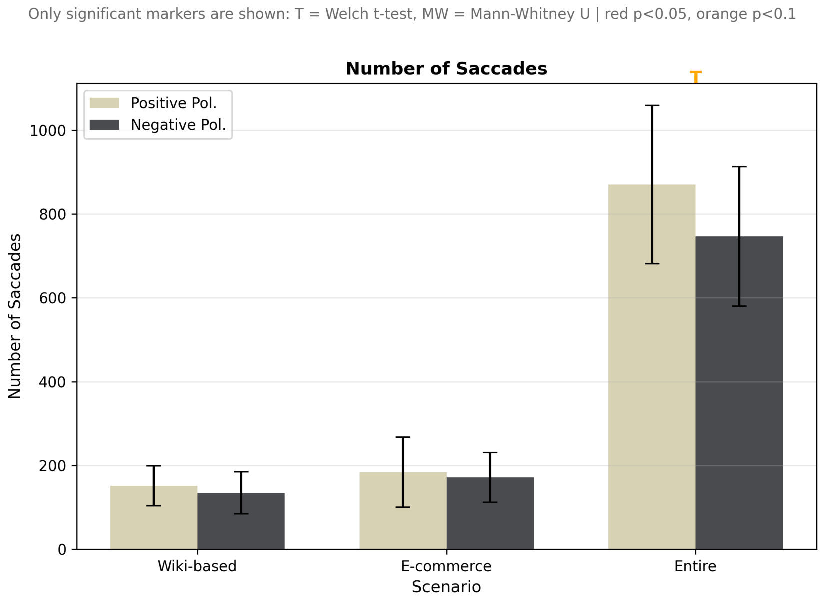

Figure 10.8: Number of Saccades for each Display Polarity and TOI.

Table 10.16: Statistics of Number of Saccades

TOI	W_{PP}	W_{NP}	T-Student	Mann-Whitney
Wiki-Based	0.9486	0.2609	0.3377	-
E-Commerce	0.1768	**0.0185**	-	0.3797
Entire Exp.	0.1577	0.6058	**0.0582**	-

10.2 Questionnaire Results

This section will provide an insight of the results obtained through the Google Forms Questionnaire. The results will be shown along with an ID, this ID corresponds to the first letter of the subsection name in the section 8.2.4 followed by the number of the question. For example, the first question of the Wiki-Based Scene will be identified as W1, the second question of the E-Commerce Scene will be identified as E2 and so on.

10.2.1 Reading Comprehension Questions

Wiki-Based

Overall results seen in table 10.17 show that the percentage of correct answers is higher in negative polarity than in positive polarity in every question but in $W3$ where the differences are pretty steep.

Table 10.17: Basic data per question of the Wiki-Based Scene

Question ID	Correct Answers	Wrong Answers	PP Correct(%)	NP Correct(%)
W1	15	19	29.41	58.82
W2	21	13	41.18	82.36
W3	20	14	70.59	47.06
W4	24	10	64.71	76.47

E-Commerce

E-Commerce results in table 10.18 show a pretty even distribution of correct answers between both polarities. The question $E3$ is surprisingly a bit lower in negative polarity as this question was pretty simple.

Table 10.18: Basic data per question of the E-Commerce Scene

Question ID	Correct Answers	Wrong Answers	PP Correct(%)	NP Correct(%)
E1	34	0	100	100
E2	18	16	47.06	58.83
E3	28	6	88.24	76.47
E4	21	13	64.71	58.82

10.2.2 Preference Questions

The questionnaire is divided in two types of questions, categorical and numeric. The categorial questions are those where the participant chooses from a non numeric list of options. For example, In question P1 the participant had to choose between "Normal Vision" and "Corrected Vision". On the other hand, the numeric questions are those where the experimentee answers with a numerical value.

Categorical Questions

The experiment remained consistent not only in gender and age distribution as seen in chapter 7 but also in the type of vision of the participants as shown in table 10.19. This consistency results in a 53% of participants with normal vision and a 47% of participants with corrected vision in both polarities.

Table 10.19: Type of vision of the participants (Question P1)

Polarity	Normal Vision	Corrected Vision
Positive Polarity	9	8
Negative Polarity	9	8

The display polarity preference for both scenes shows a slight preference for white background with black text in the Wiki-Based scene as shown in table 10.20

Table 10.20: Wiki-Based Scene Preference (Question P2)

Experiment Polarity	Positive Polarity Preferred	Negative Polarity Preferred
Positive Polarity	11	6
Negative Polarity	9	8

The trend in the E-Commerce scene is simmilar to the Wiki-Based scene having an increase for the positive polarity as seen in table 10.21.

Table 10.21: E-Commerce Scene Preference (Question P3)

Experiment Polarity	Positive Polarity Preferred	Negative Polarity Preferred
Positive Polarity	11	6
Negative Polarity	12	5

Numeric Questions

Regardin the numeric questions, the results in table 10.22 should be read along with the questions in chapter 8.2.4 to understand every question. Overall, the results appeared to be pretty even between polarities.

Table 10.22: Numeric Preference Descriptive Statistics

Question ID	Display Polarity	Min.	Mean	Std. Deviation	Max.
P4	Positive	1	3.29	1.36	5
	Negative	1	3.65	1.41	5
P5	Positive	1	2.82	1.51	5
	Negative	1	2.76	1.48	5
P6	Positive	3	4.00	0.71	5
	Negative	2	4.24	0.90	5
P7	Positive	2	3.71	0.85	5
	Negative	2	3.65	0.79	5
P8	Positive	2	3.76	0.75	5
	Negative	2	3.88	0.93	5
P9	Positive	3	4.00	0.87	5
	Negative	2	3.59	0.87	5
P10	Positive	2	3.47	0.94	5
	Negative	2	3.47	1.07	5
P11	Positive	1	3.41	1.37	5
	Negative	1	3.35	1.00	5
P12	Positive	3	4.12	0.78	5
	Negative	3	4.41	0.71	5

In adition, the statistical tests in table 10.23 show no significant results with all p-values being pretty far from the significance level. It is also important to note how none of the questions followed a normal distribution, this is probably related to the likert scale used in the questionnaire.

Table 10.23: Numeric Preference Statistical Tests

Question ID	**W_{PP}**	**W_{NP}**	**T-Student**	**Mann-Whitney**
P4	**0.014450**	**0.008111**	-	0.443897
P5	**0.018117**	**0.027799**	-	0.943579
P6	**0.003247**	**0.001766**	-	0.266670
P7	**0.033180**	**0.022905**	-	0.867835
P8	**0.006383**	**0.024222**	-	0.698056
P9	**0.001513**	**0.029538**	-	0.230296
P10	**0.039537**	**0.017035**	-	0.928482
P11	**0.047147**	**0.045448**	-	0.815614
P12	**0.003016**	**0.000501**	-	0.263356

Chapter 11

Discussion

11.1 Wiki-Based

11.2 E-Commerce

Chapter 12

Conclusion and Future Work

12.1 Lessons Learned Report

Bibliography

[1] Dark Reader. *Dark Reader Browser Extension*. URL: https://darkreader.org/.

[2] Institute of Electrical and Electronics Engineers. *IEEE Xplore Digital Library*. URL: https://ieeexplore.ieee.org/Xplore/home.jsp.

[3] C M Nadeem Faisal et al. "Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture". In: (2016). DOI: 10.1109/THMS.2016.2620901. URL: http://ieeexplore.ieee.org..

[4] Richard H. Hall and Patrick Hanna. "The impact of web page text-background colour combinations on readability, retention, aesthetics and behavioural intention". In: *Behaviour and Information Technology*. Vol. 23. May 2004, pp. 183–195. DOI: 10.1080/01449290410001669932.

[5] J.M. Redendo. *Modelo para Trabajos de Fin de Grado o Master de la Escuela de Informática de Oviedo*. URL: https://www.researchgate.net/publication/327882831_Documentos-modelo_para_Trabajos_de_Fin_de_GradoMaster_de_la_Escuela_de_Informatica_de_Oviedo.

[6] KLOSTEIN. *Mochila Viaje Cabina Avion*. URL: https://amzn.eu/d/02PDoAv5.

[7] H. B. Mann and D. R. Whitney. "On a Test of Whether One of Two Random Variables is Stochastically Larger than the Other". In: *The Annals of Mathematical Statistics* 18.1 (1947), pp. 50–60.

[8] Matplotlib Developers. *Matplotlib: Python Plotting Library*. URL: https://matplotlib.org/.

[9] Nurulain Muhamad and Nurliyana Mokhtar. "The Effect of Display Polarity on Reading Speed and Reading Error Among Young Adults". In: *Journal of Health Science and Medical Research* 42 (6 Oct. 2024), e20241095. ISSN: 2586-9981. DOI: 10.31584/jhsmr.20241095. URL: https://www.jhsmr.org/index.php/jhsmr/article/view/1095.

[10] NumPy Developers. *NumPy: Fundamental Package for Scientific Computing with Python*. URL: https://numpy.org/.

[11] Pandas Development Team. *Pandas: Python Data Analysis Library*. URL: https://pandas.pydata.org/.

[12] Alex Poole and Linden J. Ball. "Eye tracking in human-computer interaction and usability research: Current status and future prospects". In: *Encyclopedia of Human Computer Interaction*. Ed. by Claude Ghaoui. Idea Group Reference, 2006, pp. 211–219. DOI: 10.4018/978-1-59140-562-7.ch034.

[13] SciPy Developers. *SciPy: Scientific Computing Library for Python*. URL: https://scipy.org/.

[14] Scopus. *Scopus Database*. URL: https://www.scopus.com/.

[15] S. S. Shapiro and M. B. Wilk. "An analysis of variance test for normality (complete samples)". In: *Biometrika* 52.3/4 (1965), pp. 591–611.

[16] Student. "The Probable Error of a Mean". In: *Biometrika* 6.1 (1908), pp. 1–25.

[17] Tobii. *How do eye trackers work?* URL: https://www.tobii.com/resource-center/learn-articles/how-do-eye-trackers-work.

[18] Tobii. *Tobii Pro Lab Software*. URL: https://www.tobii.com/es/products/software/behavior-research-software/tobii-pro-lab.

[19] Tobii. *Tobii Pro Nano Eye Tracker*. URL: https://www.tobii.com/es/products/discontinued/tobii-pro-nano.

[20] Ru Wang et al. "Understanding how low vision people read using eye tracking". In: *Proceedings of the ACM on Human-Computer Interaction* (2023). DOI: 10.1145/3544548.3581213. URL: https://dl.acm.org/doi/full/10.1145/3544548.3581213.

[21] Wikipedia Contributors. *History of the Graphical User Interface*. URL: https://en.wikipedia.org/wiki/History_of_the_graphical_user_interface.

[22] Wikipedia Contributors. *Solanum Tuberosum*. URL: https://es.wikipedia.org/wiki/Solanum_tuberosum.

[23] Xiaojiao Xie et al. "Study on the effects of display color mode and luminance contrast on visual fatigue". In: *IEEE Access* 9 (2021), pp. 35915–35923. ISSN: 21693536. DOI: 10.1109/ACCESS.2021.3061770.

Appendix A

Project Management

A.1 Planning

A.1.1 Stakeholders

This project can be appealing to a wide variety of Stakeholders. Firstly, the academic community can be interested in the results to provide a new perspective of the influence of colors in users. It can also benefit the bursting internet industry where the user experience has become a key factor. Given this knowledge, the project can be interesting for:

- Psichology Researchers
- Human-Computer Interaction Researchers
- General Users
- Software Community and Students
- Private Companies that have a frontend product.
- Public Institutions

A.1.2 Organization Breakdown Structure (OBS) and Product Breakdown Structure (PBS)

The OBS of the project as seen in figure A.1 follows a hierarchical structure with three levels. Being the Project Manager the leader of this project and the one in charge of coordinating the software engineer and the Lead Researcher. Secondly, the Lead Researcher is in charge of both junior researchers who should be persons with knowledge in the field of computing and human computer interaction.

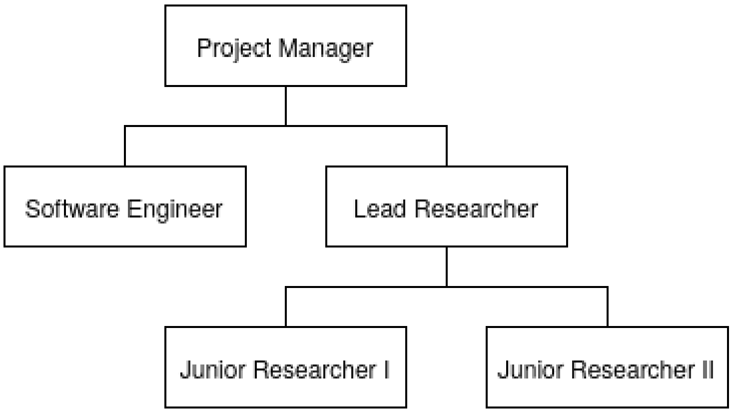

Figure A.1: Organization Breakdown Structure (OBS)

There exists two main products which compose the Product Breakdown Structure (PBS) whcih then each of them have different subtasks to develop them.

1. **Experiment:** This is the main product where all the work is focused on. It is composed of three subtasks:
   (a) **Experiment Proposal:** The State of the Art and previous planning of the experiment.
   (b) **Experiment Development:** The execution of the planned experiment.
   (c) **Data Analysis:** The processing and analysis of the obtained data.
2. **Documentation:** The final objective is to document the whole process into a scientific paper.

The PBS described have a relation to the OBS as seen in table A.1.

Table A.1: Relation between OBS and PBS

Task	Resource	Product
State of the Art	Junior Researchers, Lead Researcher	Experiment Proposal
Previous Planning	Project Manager, Lead Researcher	Experiment Proposal
Experiment Planning	Software Engineer, Lead Researcher	Experiment Development
Experiment Execution	Junior Researchers	Experiment Development
Tobii Preparations	Software Engineer	Experiment Development
Data Preprocessing	Software Engineer, Junior Researchers	Data Analysis
Data Processing	Software Engineer	Data Analysis
Statistical Study	Lead Resercher	Data Analysis
Paper Writing	Junior Researchers	Documentation
Paper Review	Lead Researcher	Documentation

A.1.3 Gantt Chart

A.1.4 Task Estimation

Table A.2: Task estimation.

WBS	Task Name	Duration (days)	Start	End
1	**Research Project**	**88.25**	**07/01/2026**	**04/05/2026**
1.1	**Previous Planning**	**13.75**	**07/01/2026**	**25/01/2026**
1.1.1	**Research Process Planning**	**1.5**	**07/01/2026**	**08/01/2026**
1.1.1.1	Needs Analysis	1	07/01/2026	08/01/2026
1.1.1.2	Organization and definition of roles	0.5	08/01/2026	08/01/2026
1.1.2	**Definition of technology needs**	**2.75**	**09/01/2026**	**12/01/2026**
1.1.2.1	Identification of technology needs	0.5	09/01/2026	09/01/2026
1.1.2.2	Selection of tools and technologies	2	09/01/2026	12/01/2026
1.1.2.3	Acquisition of chosen technologies	0.25	12/01/2026	12/01/2026
1.1.3	**Project Planning**	**9.5**	**12/01/2026**	**25/01/2026**
1.1.3.1	OBS	0.5	12/01/2026	13/01/2026
1.1.3.2	WBS	3	13/01/2026	17/01/2026
1.1.3.3	Risk Management	3	17/01/2026	21/01/2026
1.1.3.4	Budget Creation	3	21/01/2026	25/01/2026
1.2	**State of the art**	**22**	**25/01/2026**	**23/02/2026**
1.2.1	Search related articles	11	25/01/2026	08/02/2026
1.2.2	First bake	5	09/02/2026	15/02/2026
1.2.3	Second and final bake	5	15/02/2026	22/02/2026
1.2.4	Final presentation	1	22/02/2026	23/02/2026
1.3	**Experiment**	**43**	**23/02/2026**	**21/04/2026**
1.3.1	**Experiment proposal**	**7.5**	**23/02/2026**	**05/03/2026**
1.3.1.1	In depth experiment creation	5	23/02/2026	02/03/2026
1.3.1.2	Webpage selection	1	02/03/2026	03/03/2026
1.3.1.3	Creation of the form	0.5	03/03/2026	04/03/2026

Continued on next page

WBS	Task Name	Duration (days)	Start	End
1.3.1.4	Tobii pro lab configuration	1	04/03/2026	05/03/2026
1.3.2	**Experiment development**	**12**	**05/03/2026**	**21/03/2026**
1.3.2.1	Book the desired place	1	05/03/2026	06/03/2026
1.3.2.2	Preparation of material	1	07/03/2026	08/03/2026
1.3.2.3	Data Retrieval	10	08/03/2026	21/03/2026
1.3.3	**Data analysis**	**23.5**	**21/03/2026**	**21/04/2026**
1.3.3.1	**Data preprocessing**	**8**	**21/03/2026**	**01/04/2026**
1.3.3.1.1	Tobii pro lab AOI creation	4	21/03/2026	26/03/2026
1.3.3.1.2	Data download	1	27/03/2026	28/03/2026
1.3.3.1.3	Preprocesing Implementation	3	28/03/2026	01/04/2026
1.3.3.2	Data processing	6	01/04/2026	09/04/2026
1.3.3.3	Statistical study	4	09/04/2026	14/04/2026
1.3.3.4	Collect conclussions and results	5	14/04/2026	21/04/2026
1.3.3.5	Final meeting	0.5	21/04/2026	21/04/2026
1.4	**Documentation**	**6**	**22/04/2026**	**29/04/2026**
1.4.1	Abstract writing	0.5	22/04/2026	22/04/2026
1.4.2	Introduction writing	0.5	22/04/2026	23/04/2026
1.4.3	Literature writing	0.5	23/04/2026	23/04/2026
1.4.4	Method writing	1	24/04/2026	25/04/2026
1.4.5	Results and discussion writing	2	25/04/2026	27/04/2026
1.4.6	Conclusion writing	1	28/04/2026	29/04/2026
1.4.7	Reference review	0.5	29/04/2026	29/04/2026
1.5	**Project Closure**	**3.5**	**30/04/2026**	**04/05/2026**
1.5.1	Final documentation review	3	30/04/2026	03/05/2026
1.5.2	Paper and project handout	0.5	04/05/2026	04/05/2026

A.2 Risk Identification and Management

Risk identification is a key phase of the project as it can help acknowledge possible negative effects on the project. It is important to have a strategy to face such events in order to make them affect the least possible the outcome of the project.

Table A.3: Risks List

Risk ID	Risk Name
1	Few Experimental Samples
2	Statistical Analysis Imprecision
3	Loss of Partial Work
4	Justified Leave of Worker
5	Lack of Teamwork
6	Tobii Measurements Drifts
7	Privacy Violations
8	Timeline Delay
9	Publisher Rejection

A.2.1 Risk Details

Few Experimental Samples

There is not enough sample for the test, this can lead to inconsistent data and problems when doing the processing and publishing the paper.

Table A.4: Few experimental samples impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	Low	Medium	Critic	0,45

Statistical Analysis Imprecision

When analyzing the processed data if the statistic analysis is imprecise it can lead to wrong conclusions. It is important for it to remain precise.

Table A.5: Statistical analysis imprecision impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Seamless	Low	Low	Medium	0,09

Loss of Partial Work

Due to data corruption or any other external agent, there's a possibility of losing the entire or part of the work accomplished.

Table A.6: Loss of partial work impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Medium	High	Medium	Low	0,17

Justified Leave of Worker

Any of the workers may need a justified leave of work according to the Spanish law.

Table A.7: Justified leave of worker impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	High	Low	Low	0,28

Lack of Teamwork

Teamwork is important as it creates a healthy environment where workers feel comfortable, however it is quite easy to have a lot of individualism in teams.

Table A.8: Lack of teamwork impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Medium	Low	High	Low	Low	0,15

Tobii Measurements Drifts

Due to light changes, the subject of the experiment looking away or other uncontrollable events, the tobii eye-tracker may drift its measurements.

Table A.9: Tobii measurements drifts impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
High	Seamless	Seamless	Seamless	Medium	0,21

Privacy Violations

This whole research gathers a lot of sensitive information and it is a key to store and manipulate the sensitive data correctly as it can lead to fines.

Table A.10: Privacy violations impact.

Probability	Impact				Impact
	Budget	Planning	Reach	Quality	
Low	Critic	Low	Seamless	Low	0,27

Timeline Delay

Research projects usually take longer than initially expected this can overextend the planning making it a real risk.

Table A.11: Timeline delay impact.

Probability	**Impact**				**Impact**
	Budget	**Planning**	**Reach**	**Quality**	
High	High	High	Medium	Low	0,39

Publisher Rejection

After finishing the experiment, gathering the data and writing the paper there's a possibility the desired scientific publishers may not be interested in the paper.

Table A.12: Publisher rejection impact.

Probability	**Impact**				**Impact**
	Budget	**Planning**	**Reach**	**Quality**	
High	Seamless	Low	High	Low	0,39

A.2.2 Risk Strategy

A risk strategy is needed in order to know what to do against every risk, the risks of A.2.1 explains them further. From the 9 identified risks, 2 will be deleted, 3 assumed and 4 mitigated.

Table A.13: Risk strategy and action plan.

ID	Risk Name	Impact	Strategy	Risk Answer
1	Few Experimental Samples	0,45	Delete	Spend more time than planned gathering samples for the experiment.
8	Timeline Delay	0,39	Mitigate	When creating the planning always adding more time than expected.
9	Publisher Rejection	0,39	Assume	It should be necessary to have a broad list of publishers.
4	Justified Leave of Worker	0,28	Assume	Work the planning around the person missing.
7	Privacy Violations	0,27	Delete	Exhaustive analysis by the prepared workers on the RGPD.
6	Tobii Measurements Drifts	0,21	Assume	Try to find a spot with balanced lighting.
3	Loss of Partial Work	0,17	Mitigate	Integrate a version control system on the work.
5	Lack of Teamwork	0,15	Mitigate	Weekly optional team buildings outside of the work period.
2	Statistical Analysis Imprecision	0,09	Mitigate	Final deep review of the statistical analysis.

A.3 Incidence Log

The incidence log is a document where incidents are described with their date. It serves as a way to understand and keep track of the incidents that happen during the project.

Table A.14: Incidence log.

Date	Incident Description	Solution
17/10/2024	Lack of availability in scientific websites	After contacting the professors, it was possible to find scientific paper searchers.
24/04/2025	Lack of participants for the experiment	Bringing known colleagues helped gathering data and creating inertia to gather more participants.
18/05/2025	Documentation and Tobii Data Loss	Version control was established and the latest available version was restored.
27/05/2025	Project delay	Due to not enough availaiblity the problem was solved by extending to next year the deadline of the project.
12/11/2025	Problems with titles and table of contents	After hard work, the documentation was moved from word to latex.
01/05/2026	Pywib early version problems	After contacting with the current mantainer and the early versions of the program, It was decided to stop the use of this hci library develeopded in Uniovi. Worht keeping an eye on in the future.

Appendix B

Project Budget

Table B.1 shows the staff costs for the project, including the gross annual salary and the annual salary cost for each role with an estimated 30% overhead. The total staff cost for the project is estimated to be 215,800.00 €per year.

Staff	**Gross Annual Salary (€)**	**Annual Salary Cost (€)**
Project Manager	42,000.00	54,600.00
Lead Researcher	42,000.00	54,600.00
Software Engineer	32,000.00	41,600.00
Junior Researcher I	25,000.00	32,500.00
Junior Researcher II	25,000.00	32,500.00
Total		215,800.00

Table B.1: Staff costs for the project

Given the table B.1 and estimating the productivity of each staff member, we can calculate the direct and indirect costs for each role. The direct costs is the percentage of the salary that is related to the productivity, while the indirect costs is the remaining percentage of the salary.

Staff	**Total (€)**	**Productivity (%)**	**Direct Cost(€)**	**Indirect Cost (%)**	**I.C. (€)**
Project Manager	54,600.00	0	-	100	54,600.00
Lead Researcher	54,600.00	70	38,220.00	30	16,380.00
Software Engineer	41,600.00	80	33,280.00	20	8,320.00
Junior Researcher I	32,500.00	80	26,000.00	20	6,500.00
Junior Researcher II	32,500.00	80	26,000.00	20	6,500.00
Total	215,800.00		123,500.00		92,300.00

Table B.2: Estimated productivity for each staff member

Along with the indirect costs calculated in table B.2, there are other indirect costs that must be taken into account these are the costs relative to the services used as seen in table B.3. The total indirect costs for the project is estimated to be 2,831.88 €.

Service	**Monthly Cost (€)**	**Yearly Cost (€)**
Cleaning	20.00	240.00
Light Consumption	70.00	840.00
Water Consumption	20.00	240.00
Heating	20.00	240.00
Internet Provider	65.99	791.88
Movility Costs	40.00	480.00
Total		2,831.88

Table B.3: Other Indirect Costs for the project

Another indirect cost is the amortization of equipment that must be acquired for the project like the eye-tracker thsi is sumarized in table B.4. The total procution cost for the project is estimated to be 358.33 €.

License/Team	**Units**	**Price (€)**	**Total Price (€)**	**Annual Cost (€)**	**Type**
Tobii pro nano	1	2,500.00	2,500.00	208.33	Amortization
Tobii Pro Lab License	1	1,800.00	1,800.00	160.00	Amortization
			Total	358.33	

Table B.4: Summary of production costs for the project

The billing needs for the project is calculated in table B.5.

Type	**Total Cost (€)**
Indirect plus direct costs	218,990.21
Desired Benefit (25%)	54,747.55
Billing Needs	273,737.77

Table B.5: Billing Needs for the project

Given a year of the project the productive hours the staff is able to reach is calculated in table B.6

Staff	Prod.(%)	Hours/Year	Prod. hours/year
Project Manager	0	1784	0
Lead Researcher	70	1784	1249
Software Engineer	80	1784	1427.2
Junior Researcher I	80	1784	1427.2
Junior Researcher II	80	1784	1427.2
		Total	5530.4

Table B.6: Schedule for the project

Since the productive hours habe been estimated, the price per hour for each staff member is calculated accordingly in table B.7. Please remember that the total billing is for a whole productive year.

Staff	Price/h (€)	Price/h no benefit (€)	Prod. Hours	Billing
Project Manager	-	-	0	0
Lead Researcher	70.00	56.00	1248.8	87,416.00
Software Engineer	50.00	40.00	1427.2	71,360.00
Junior Researcher I	41.25	33.00	1427.2	58,872.00
Junior Researcher II	41.25	33.00	1427.2	58,872.00
			Total	276,520.00

Table B.7: Price per hour for each staff member

In summary, the total cost for the project in a whole productive year is calculated in table B.8. Where the total margin of costs and billing is $2,782.23$€ or 1.02% of the total billing needs. This margin proves that the project is financially viable.

Concept	Cost (€)
Total Direct Costs	123,500.00
Total Indirect Costs	98,490.21
Sum of Direct and Indirect Costs	218,990.21
Desired Benefit (25%)	54,747.55
Total Cost(Sum of costs + benefit)	273,737.77
Possible Billing by productivity and price per hour	276,520.00

Table B.8: Summary of project costs

Given the information of tables above, the total budget for the client project that will be further explained in appendix C is calculated in table B.9.

The cost for the client is calculated by adding the results of multiplying the estimated cost by the percentage to share with the clients. This percentage is obtained by dividing the sum of the benefits and the indirect cost of "PRevious Planning and State of the Art" by the total cost of the project minus these indirect costs.

Part	**Cost (€)**	**Cost for client (€)**
1. Previous Planning and State of the Art	6,622.50	
1.1 Previous Planning	1,005.00	
1.2 State of the Art	5,617.50	
2. Experiment	13,588.75	23,991.57
2.1 Experiment Proposal	2,943.75	5,197.33
2.2 Experiment Development	2,722.50	4,806.70
2.3 Experiment Data Analysis	7,922.50	13,987.54
3. Documentation for the paper	3,041.25	5,369.47
Total	23,252.50	29,361.04
Total with Benefits	29,065.63	
Contingency	295.42	
Benefits	6,108.54	
Percentage to share with client	76.55%	

Table B.9: Final budget of the project

Appendix C

Client Budget

The following table C.1 shows the initial budget estimation of the project for the client. These parts will be further detailed in the following sections Preparations C.1, Experiment C.2, and Documentation C.3.

Part	**Total Cost(€)**
1. Experiment	23,991.57
1.1 Experiment Proposal	5,197.33
1.2 Experiment Development	4,806.70
1.3 Data Analysis	13,987.54
2. Documentation	5,369.47
Total	29,361.04

Table C.1: Initial budget estimation for the project

C.1 Preparations

This part of the budget includes the planning ahead that is not included in the clients budget directly, but it is necessary to be able to execute the project. The fact of this part not existing in the client budget is one of the reasons why the future parts total cost won't be equal to the ones appearing in table C.1. In order to see how these indirect costs are distributed please refer to appendix B.

Table C.2: Preparations part

Task Name	**Qty. (hours)**	**Cost (€)**	**Sub. 3 (€)**	**Sub. 2 (€)**	**Sub. 1 (€)**	**Total (€)**
1. Previous Planning						1,005.00
1.1 Research Process Planning					-	
1.1.1 Needs Analysis				-		
1.1.1.1 Project Manager	6.0	-	-			
1.1.2 Organization and definition of roles						
1.1.2.1 Project Manager	3.0	-	-			
1.2 Definition of technology needs					1,005.00	
1.2.1 Identification of technology needs				210.00		
1.2.1.1 Lead Researcher	3.0	70.00	210.00			
1.2.2 Selection of tools and technologies				720.00		
1.2.2.1 Lead Researcher	6.0	70.00	420.00			
1.2.2.2 Software Engineer	6.0	50.00	300.00			
1.2.3 Acquisition of technologies				75.00		
1.2.3.1 Software Engineer	1.5	50.00	75.00			
1.3 Project Planning					-	

Continued on next page

Task Name	**Qty. (hours)**	**Cost (€)**	**Sub. 3 (€)**	**Sub. 2 (€)**	**Sub. 1 (€)**	**Total (€)**
1.3.1 OBS				-		
1.3.1.1 Project Manager	3.0	-	-			
1.3.2 WBS				-		
1.3.2.1 Project Manager	18.0	-	-			
1.3.3 Risk Management				-		
1.3.3.1 Project Manager	18.0	-	-			
1.3.4 Budget				-		
1.3.4.1 Project Manager	18.0	-	-			
2. State of the Art						5,617.50
2.1 Search and Analysis					2,722.50	
2.1.1 Junior Researcher I	33.0	41.25		1,361.25		
2.1.2 Junior Researcher II	33.0	41.25		1,361.25		
2.2 First Bake					1,237.50	
2.2.1 Junior Researcher I	15.0	41.25		618.75		
2.2.2 Junior Researcher II	15.0	41.25		618.75		
2.3 Second and Final Bake					1,237.50	
2.3.1 Junior Researcher I	15.0	41.25		618.75		
2.3.2 Junior Researcher II	15.0	41.25		618.75		
2.4 Final Presentation					420.00	
2.4.1 Lead Researcher	6.0	70.00		420.00		
					Total	6,622.50

C.2 Experiment

This part of the budget includes all the related costs and tasks of the experiment preparation, execution and processing. It is divided exactly in those three subparts that were just mentioned and it represents the most expensive and time consuming part of the project, being the data analysis the most one of them three, as seen in the table C.3. Please see the table in the followin page as the table is too big to fit in one page so it's landascaped.

Table C.3: Experiment part

Task Name	**Qty. (hours)**	**Cost (€)**	**Sub. 3 (€)**	**Sub. 2 (€)**	**Sub. 1 (€)**	**Total (€)**
1. Experiment Proposal						2,943.75
1.1 In depth experiment creation					2,100.00	
1.1.1 Lead Researcher	30.00	70.00		2,100.00		
1.2 Webpage selection					300.00	
1.2.1 Software Engineer	6.0	50.00		300.00		
1.3 Creation of the form					123.75	
1.3.1 Junior Researcher II	3.0	41.25		123.75		
1.4 Tobii Pro Lab configuration					420.00	
1.4.1 Lead Researcher	6.0	70.00		420.00		
2. Experiment Development						2,722.50
2.1 Book the desired place					247.50	
2.1.1 Junior Researcher I	6.0	41.25		247.50		
2.2 Prep. of material					247.50	
2.2.1 Junior Researcher II	6.0	41.25		247.50		
2.3 Data Retrieval					2,475.00	
2.3.1 Junior Researcher I	30.0	41.25		1,237.50		

Continued on next page

Task Name	Qty. (hours)	Cost (€)	Sub. 3 (€)	Sub. 2 (€)	Sub. 1 (€)	Total (€)
2.3.2 Junior Researcher II	30.0	41.25		1,237.50		
3. Data Analysis						7,922.50
3.1 Data Preprocessing					2,190.00	
3.1.1 Tobii Pro Lab AOI Creation				990.00		
3.1.1.1 Junior Researcher I	24.0	41.25	990.00			
3.1.2 Data Download				300.00		
3.1.2.1 Software Engineer	6.0	50.00	300.00			
3.1.3 Preprocessing Implementation				900.00		
3.1.3.1 Software Engineer	18.0	50.00	900.00			
3.2 Data Processing					1,800.00	
3.2.1 Software Engineer	36.0	50.00		1,800.00		
3.3 Statistical Study					1,680.00	
3.3.1 Lead Researcher	24.0	70.00		1,680.00		
3.4 Collect Conclusions and results					2,100.00	
3.4.1 Lead Researcher	30.0	70.00		2,100.00		
3.5 Final Meeting					152.50	
3.5.1 Junior Researcher I	1.0	41.25		41.25		
3.5.2 Junior Researcher II	1.0	41.25		41.25		
3.5.3 Lead Researcher	1.0	70.00		70.00		
					Total	13,588.75

C.3 Documentation

The Documentation part is centered around the idea of writign a scientific paper with the results of the experiment. This part is the cheapest one of the main three, hwoever it is still a key factor for the success of the project.

Task Name	**Qty. (hours)**	**Cost (€)**	**Sub. 2 (€)**	**Sub. 1 (€)**	**Total (€)**
1. Documentation					3,041.25
1.1 Abstract Writing				123.75	
1.1.1 Junior Researcher I	3.0	41.25	123.75		
1.2 Introduction Writing				123.75	
1.2.1 Junior Researcher II	3.0	41.25	123.75		
1.3 Literature Writing				123.75	
1.3.1 Junior Researcher I	3.0	41.25	123.75		
1.4 Method Writing				247.50	
1.4.1 Junior Researcher II	6.0	41.25	247.50		
1.5 Results and Discussion Writing				495.00	
1.5.1 Junior Researcher I	12.0	41.25	495.00		
1.6 Conclusion Writing				247.50	
1.6.1 Junior Researcher II	6.0	41.25	247.50		
1.7 Reference Review				420.00	
1.7.1 Lead Researcher	6.0	70.00	420.00		
1.8 Final Review				1,260.00	
1.8.1 Lead Researcher	18.0	70.00	1,260.00		
				Total	3,041.25

Table C.5: Documentation part